Name:____ **Date:** ____________

Problem 1. *Order the following functions from slowest-growing to fastest-growing:* $x^2, x, \ln(x), e^x, x^x$.

Problem 2. *Compute the following limits (hint: your answer to problem 1 may be helpful):*

***a*)** $\displaystyle\lim_{x\to\infty} \frac{x^2}{\ln(x)}$

***b*)** $\displaystyle\lim_{x\to\infty} \frac{\ln(x)}{x}$

***c*)** $\displaystyle\lim_{x\to\infty} \frac{e^x}{x^x}$

***d*)** $\displaystyle\lim_{x\to\infty} \frac{x^x}{x^2}$

***e*)** $\displaystyle\lim_{x\to\infty} \frac{x^2}{e^x}$

***f*)** $\displaystyle\lim_{x\to\infty} \frac{x}{x^2}$

***g*)** $\displaystyle\lim_{x\to\infty} \frac{\ln(x)}{e^x}$

Problem 3. *Compute the limits of the following rational functions:*

a) $\lim\limits_{x\to\infty} \dfrac{x^3 - 6x + 15}{3x^2 - 12}$

b) $\lim\limits_{x\to\infty} \dfrac{3x^5 - 16x^3 + 12x - 3}{7x^5 - 46x^2}$

c) $\lim\limits_{x\to\infty} \dfrac{x - 5}{x^2 + 3x - 2}$

d) $\lim\limits_{x\to\infty} \dfrac{2x^2 + 11}{x^3 + 3x + 11}$

e) $\lim\limits_{x\to\infty} \dfrac{15x^7 - 8x^6 + 13x^5 - 4x^4 + x^3 + 17x - 5}{3x^7 + 12x^4 - 3x^2 + 2}$

f) $\lim\limits_{x\to\infty} \dfrac{ax + b}{cx + d}$

g) $\lim\limits_{x\to\infty} \dfrac{ax + b}{cx^2 + dx + e}$

h) $\lim\limits_{x\to\infty} \dfrac{ax^3 - b}{cx^3 + dx - 5}$